Ryerson University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title:	Electric Networks
Course Number:	ELE 302
Semester/Year (e.g.F2016)	F2026

Instructor:	Dr. Andy Ye

Assignment/Lab Number:	2
Assignment/Lab Title:	The Step Response of Second-Order Circuits

Submission Date:	October 17, 2026
Due Date:	October 15, 2026

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a “0” on the work, an “F” in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: http://www.ryerson.ca/senate/current/pol60.pdf

5.0 IN-LAB IMPEMENTATION & MEASUREMENTS (5 marks in total):

Part I: The Step Response of a Second-Order Bandpass Circuit

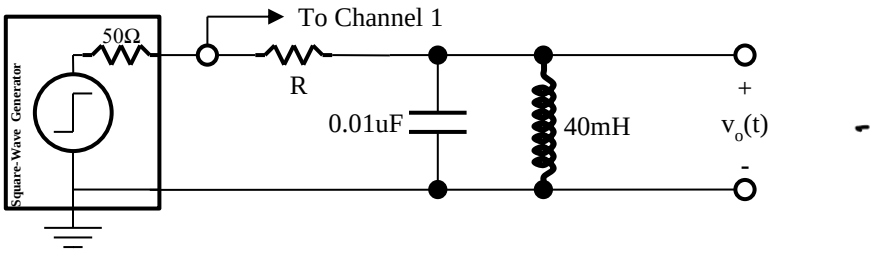


Figure 5.0: Second-Order Bandpass Filter

(a) Step 1: Connect Channel (1) of the oscilloscope to display the open-circuit voltage $v_s(t)$ of the function generator. Set the following:

- Trigger: source→Channel (1), and slope→rising.

Adjust the controls of the function generator to provide a square-wave signal $v_s(t)$ with a peak-to-peak value of 10V and DC offset of 5V at a frequency of 20Hz.
For ENG301, 302, 303 (FG): Waveform → Square, Frequency → 20 Hz, Amplitude → 5 Vpp*, Offset → 2.5 V*

*Due to FG matching impedance (EDU33212A)

For ENG402 (FG): Directly connected it with Oscilloscope, adjust AMPL knob to get 5 Volt peak (Vp) and adjust the OFFSET knob (you may need to pull it out), to get +5V DC off set. *Function Generator GFG8216A

Next, set the following:

- Channel (2): Vertical-position→one division above the bottom of the screen, coupling→dc, and V/div→1V.
- Time: Time/div→50μs when $v_s(t)$ is rising.
- Blank off Channel (1).

(b) Step 2: Construct the circuit shown in **Figure 5.0**. Set R=3.4kΩ. Connect Channel (2) to display the step response $v_o(t)$. Plot $v_o(t)$ on **Graph 1.0.**

(c) Step 3: Use the cursors on the oscilloscope to measure points on the $v_o(t)$ display to calculate the parameters (σ and ω) that characterize the step response as:

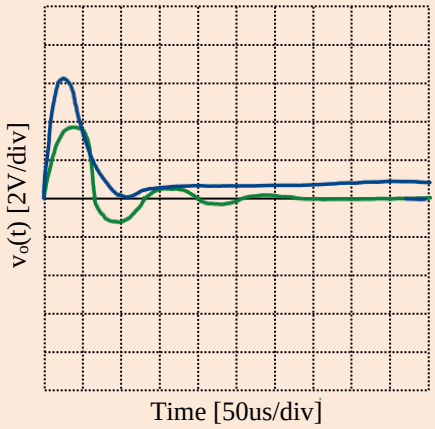
$$\mathbf{v_o}(t) = A\boldsymbol{e}^{-\boldsymbol{\sigma t}}\sin(\omega t)$$

(d) Step 4: Record your results in **Table 1.0.**

(e) Step 5: Demonstrate the correct operation of your setup to your TA. (1 mark)

(f) Step 6: Set R=1.4kΩ, and repeat as in Step 2 and Step 3. Record your results in **Table 1.0.**

(g) Step 7: Set R=500Ω, and repeat as in Step 2.



Graph 1.0 (0.5 marks)

Table 1.0 (0.5 marks)

R (KΩ)	T	$v_o(T/4)$	$v_o(3T/4)$	σ	ω
3.4	127 μs	3.63 V	-1.04 V	19 685.2	49473.9
1.4	136μs	5.51 V	300 mV	42 802	46 199.4

Part II: The Step Response of a Second-Order Lowpass Circuit

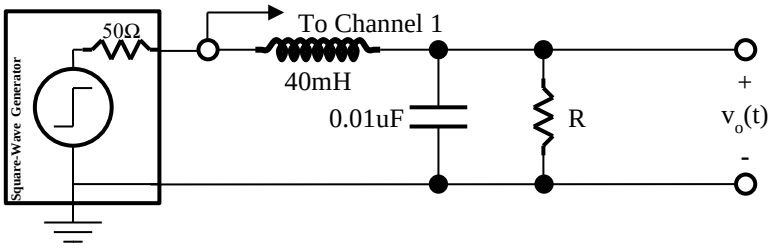


Figure 6.0: Second-Order Lowpass Filter

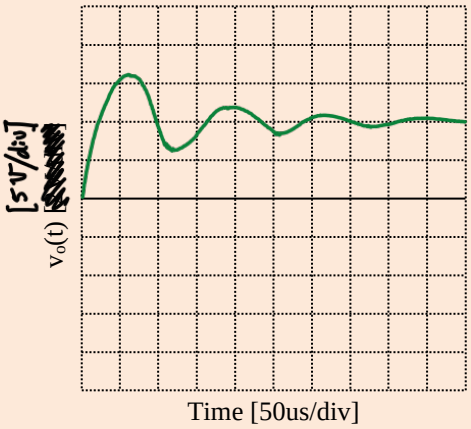
(h) Step 8: Connect the circuit shown in **Figure 6.0.** Set R=10kΩ. Use Channel (2) to display the step response v_o(t). Plot v_o(t) on **Graph 2.0.**

(i) Step 9: Use the cursors on the oscilloscope to measure points on the v_o(t) display to calculate (and record in **Table 2.0**) the parameters (σ and ω) that characterize the step response as:

$$\mathbf{v_o}(t) = B + Ae^{-\sigma t}\cos(\omega t + \theta)$$

(j) Step 10: Set R=1.4kΩ, and repeat as in Step 8.

(k) Step 11: Set R=1kΩ, and repeat as in Step 8.



Graph 2.0 (0.5 marks)

Table 2.0 (0.5 marks)

R (KΩ)	T	$v_o(T/4)$	$v_o(3T/4)$	σ	ω
10	127 µs	15.85	6.21	7179	99473.9

Part III: The Step Response of a Second-Order Highpass Circuit

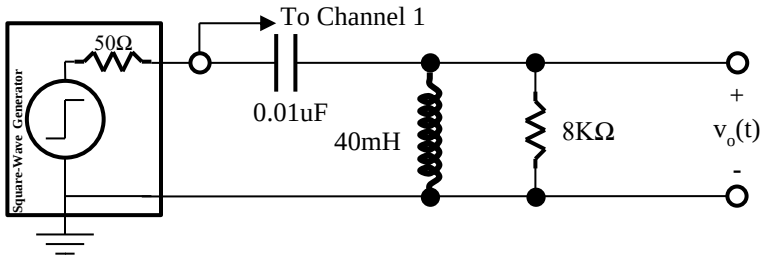


Figure 7.0: Second-Order Highpass Filter

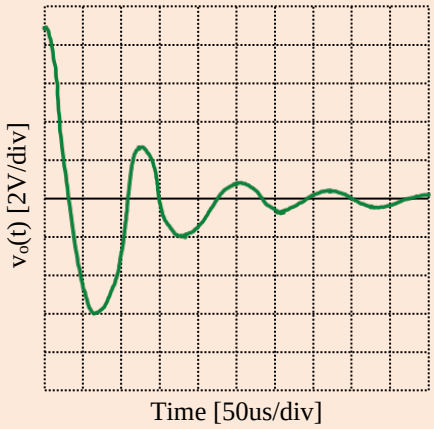
(l) Step 12: Connect the circuit shown in **Figure 7.0.** Connect Channel (2) to display the step response $v_o(t)$. Plot $v_o(t)$ on **Graph 3.0.**

(m) Step 13: Use the cursors on the oscilloscope to measure points on $v_o(t)$ display to calculate (and record in **Table 3.0**) the parameters (σ and ω) that characterize the step response as:

$$\mathbf{v_o}(t) = Ae^{-\sigma t}\cos(\omega t + \theta)$$

(n) Step 14: Demonstrate the correct operation of your setup to your TA. (1 mark)

0



Graph 3.0 (0.5 marks)

Table 3.0 (0.5 marks)

R (KΩ)	T	v_o(T/4)	v_o(3T/4)	σ	ω
8	128 μs	-6.04	3.06	10 624.8	49 087.4

6.0POST-LAB QUESTIONS (2 marks in total, 2/3 marks for each question):

(1) By examining your plots on **Graph 1.0,** answer the following:

a) What are the effects of varying the value of R on the step response of a second-order bandpass circuit?

A longer R gave a smaller σ value (19.7k vs. 42.8k) which meant the circuit is less damped. This allowed the oscillation to last longer as the decay was smaller.

A smaller R gave the opposite. A smaller σ value which meant the circuit is more damped. This lead to the graph to decay faster.

Furthermore, the frequency increases slightly as R increases and therefore the period is shorter.

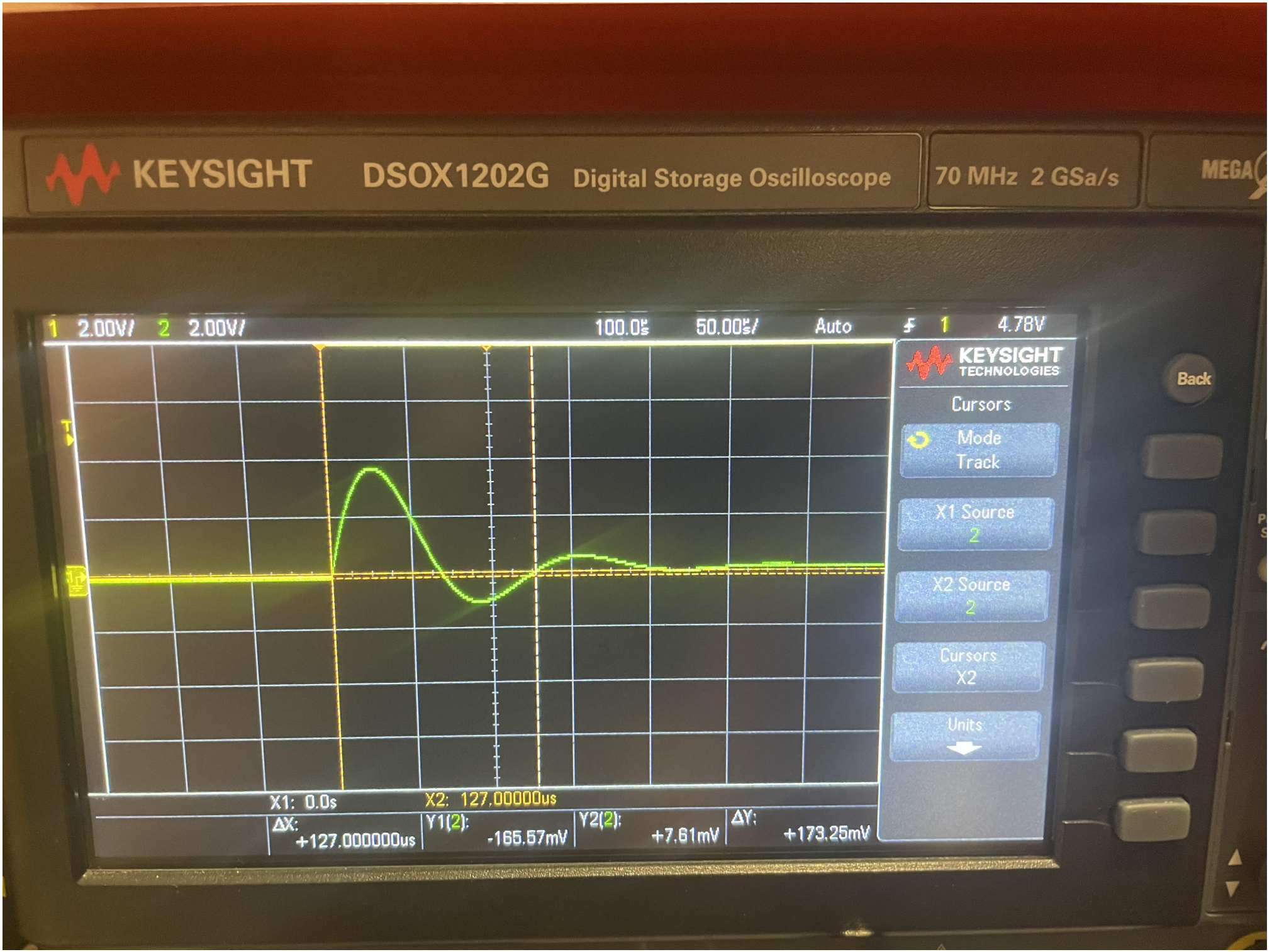
(2) By examining your plots on **Graph 2.0,** answer the following:

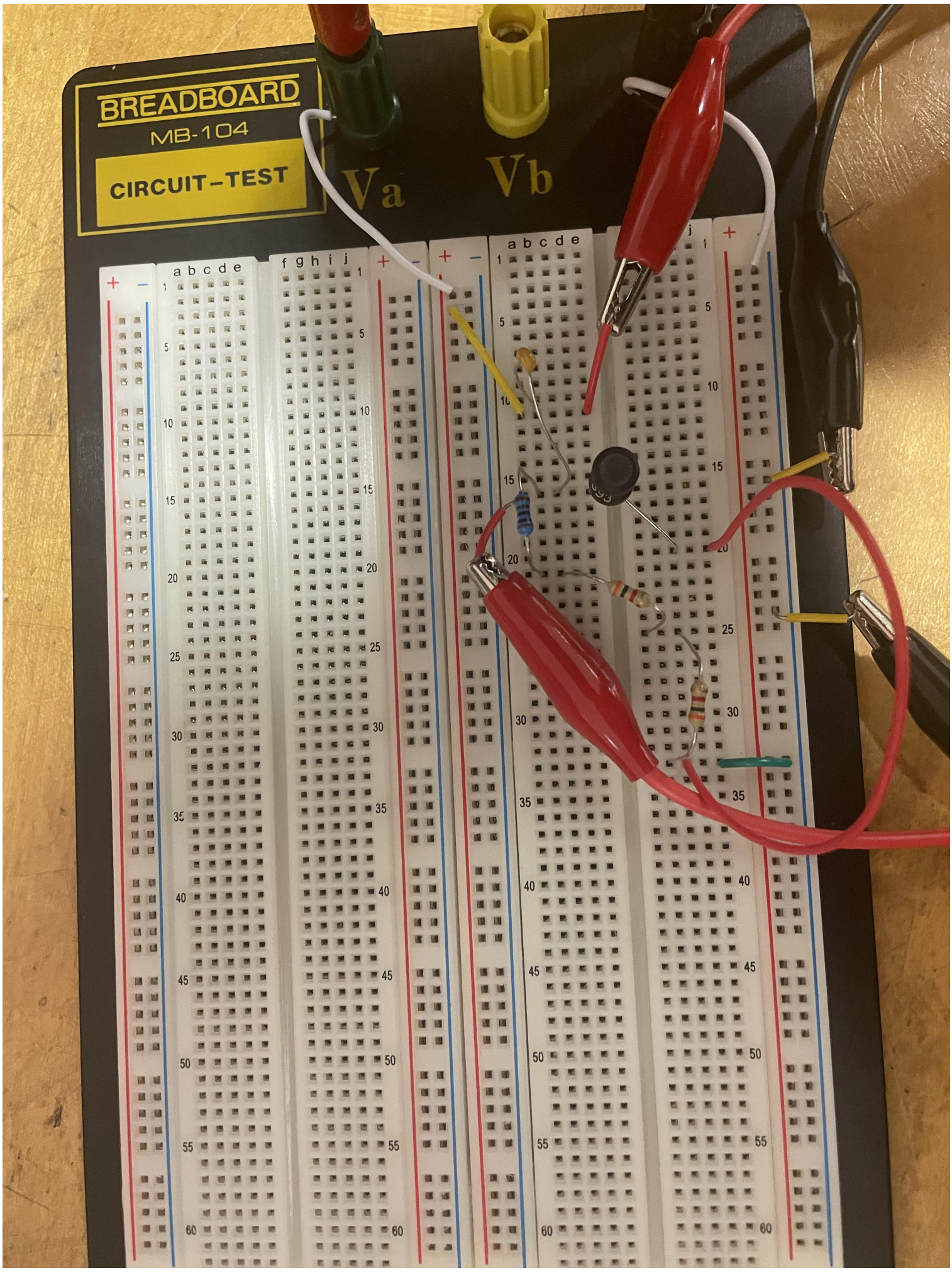
a) What are the effects of varying the value of R on the step response of a second-order lowpass circuit?

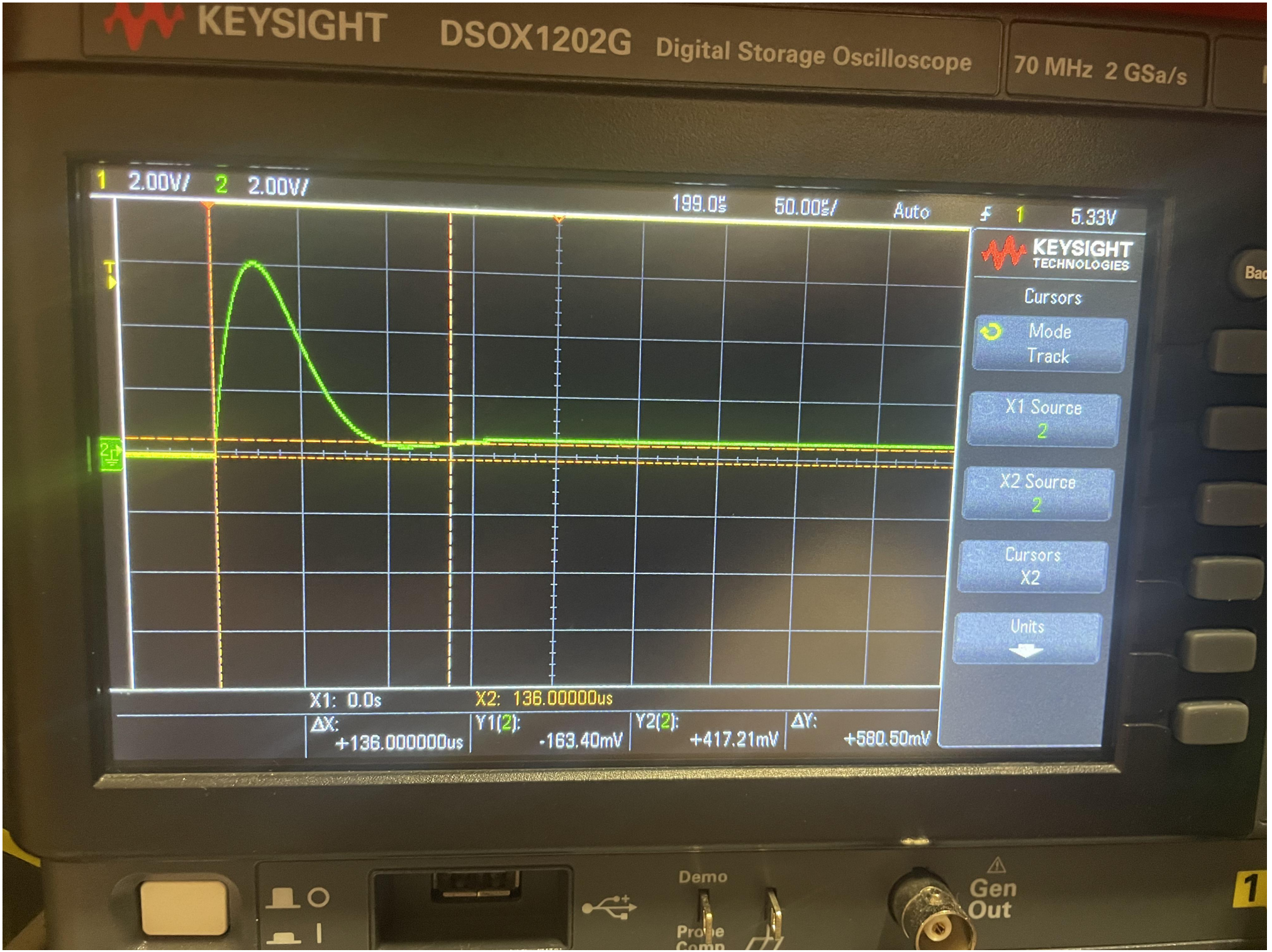
A larger R gives a smaller σ value. At 10k, the response was underdamped so it goes well past the final value (15.85V vs. 9.95V) and oscillates several times before settling. Decreasing R lowers σ which dampens the circuit more and causes the overshoot to shrink as well as shrinks the oscillation. Any resistance under around 1kΩ will be below critical dampening ($\sigma = \omega_0$) and the circuit will rise without overshoot.

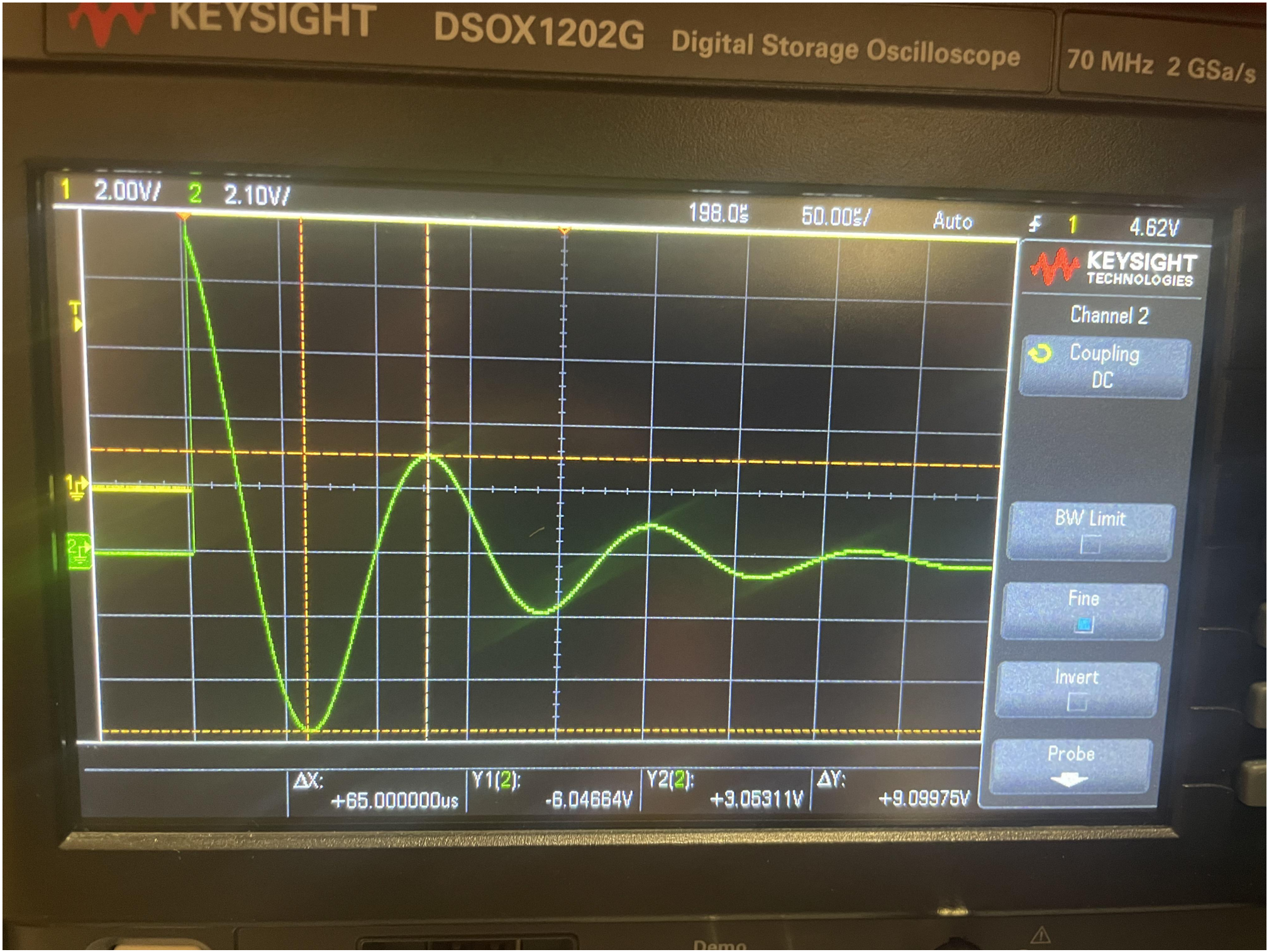
(3) Suppose that the 8kΩ-resistor is removed from the circuit in Fig (2.6), what effects will this have on the step response?

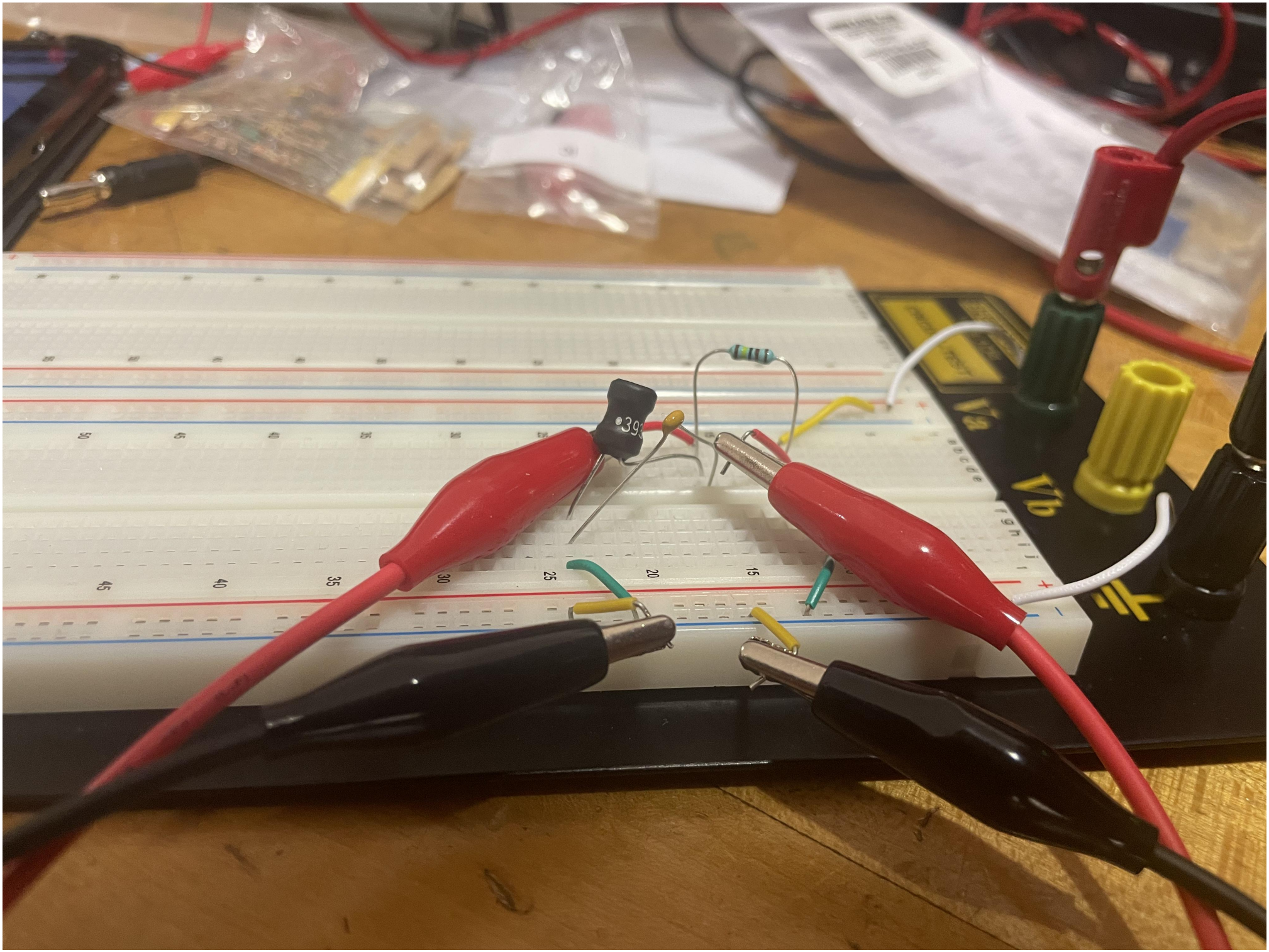
If you remove the resistor, the circuit would become a series C-L circuit with only a 50 Ω resistance providing the dampening. This will cause α to drop from about 6900/s to 625/s. This means the decay would be a lot smaller so the response lasts a lot longer and oscillates a lot more.

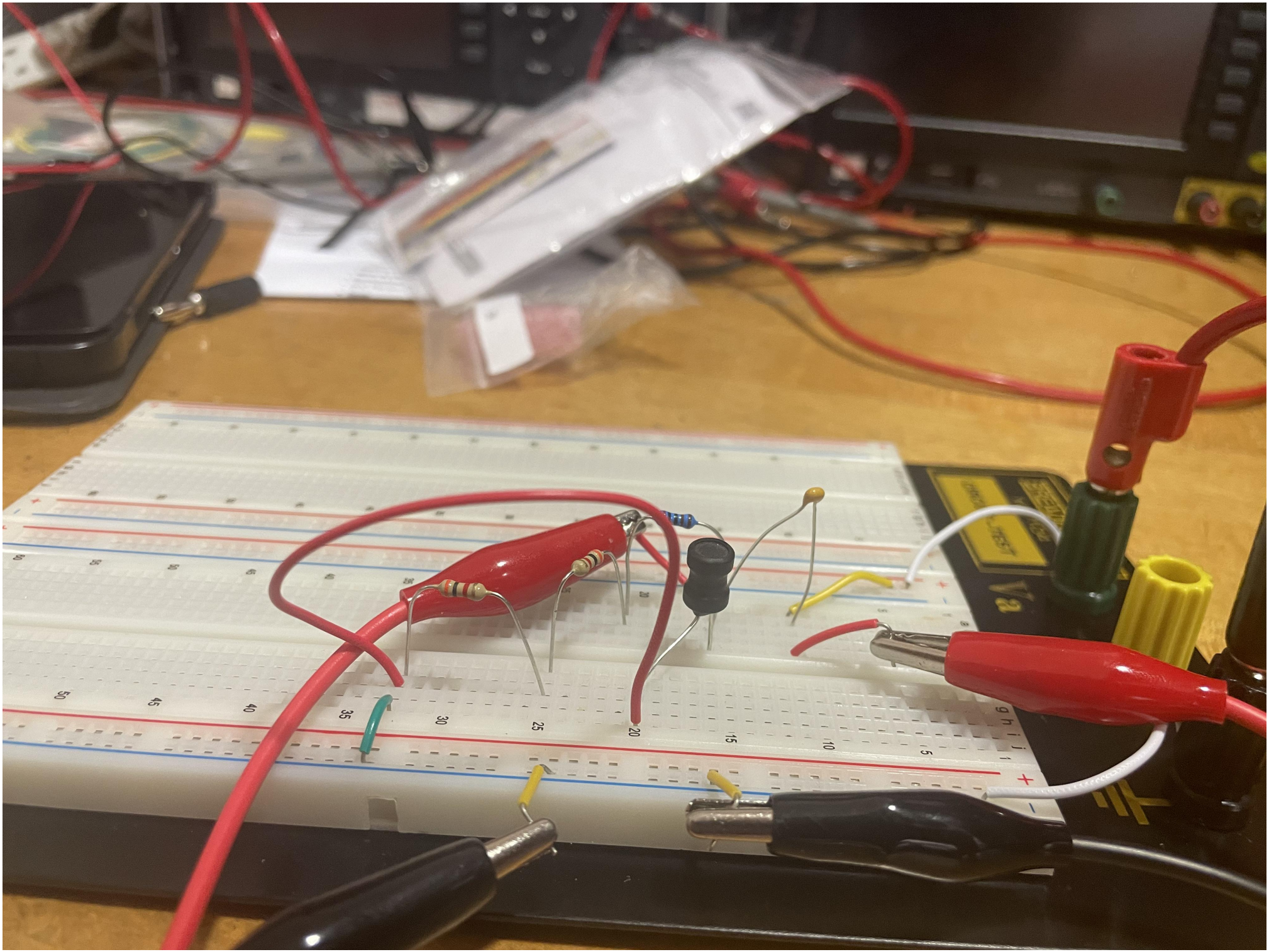

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
MEGA
1 2.00V/ 2 2.00V/
100.0ns
50.00ns/
Auto
1 4.78V
KEYSIGHT TECHNOLOGIES
Cursors
Mode
Track
X1 Source
2
X2 Source
2
Cursors
X2
Units
Back
X1: 0.0s
X2: 127.00000us
ΔX:
+127.000000us
Y1(2):
-165.57mV
Y2(2):
+7.61mV
ΔY:
+173.25mV


BREADBOARD
MB-104
CIRCUIT–TEST
Va
Vb

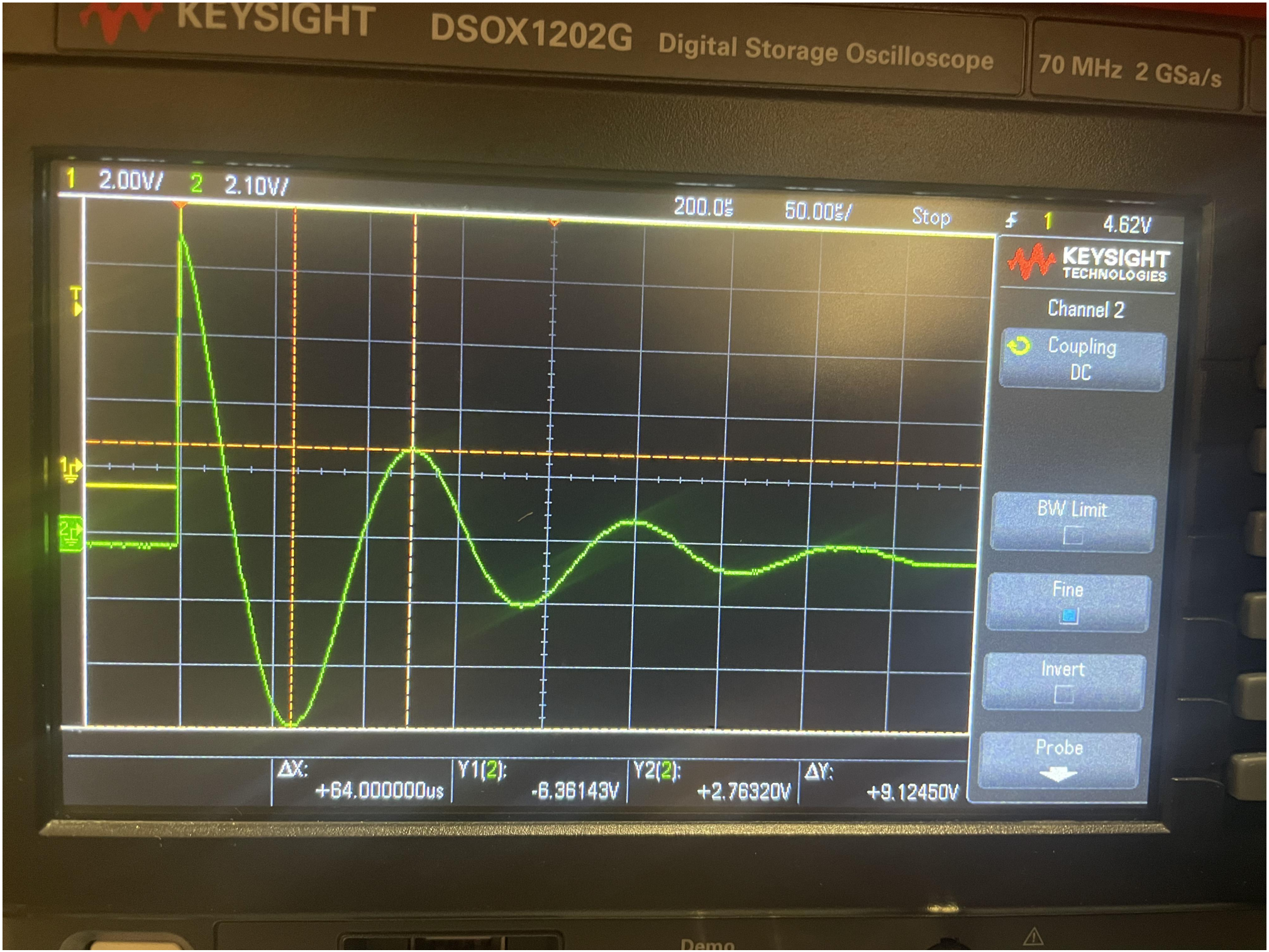

KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 2.00V/ 2 2.00V/
199.0ns
50.00ns/
Auto
1 5.33V
KEYSIGHT TECHNOLOGIES
Cursors
Mode
Track
X1 Source
2
X2 Source
2
Cursors
X2
Units
X1: 0.0s
X2: 136.00000us
ΔX:
+136.000000us
Y1(2):
-163.40mV
Y2(2):
+417.21mV
ΔY:
+580.50mV
Demo
Probe Comp
Gen Out
1


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 2.00V/ 2 2.10V/
198.0ns
50.00ns/
Auto
1 4.62V
KEYSIGHT TECHNOLOGIES
Channel 2
Coupling
DC
BW Limit
Fine
Invert
Probe
ΔX:
+65.000000us
Y1(2):
-6.04664V
Y2(2):
+3.05311V
ΔY:
+9.09975V


Va
Vb



Vb
Va
MB-104


KEYSIGHT
DSOX1202G
Digital Storage Oscilloscope
70 MHz 2 GSa/s
1 2.00V/ 2 2.10V/
200.0µs 50.00µs/ Stop
1 4.62V
KEYSIGHT TECHNOLOGIES
Channel 2
Coupling
DC
BW Limit
Fine
Invert
Probe
ΔX: +64.000000us
Y1(2): -6.36143V
Y2(2): +2.76320V
ΔY: +9.12450V